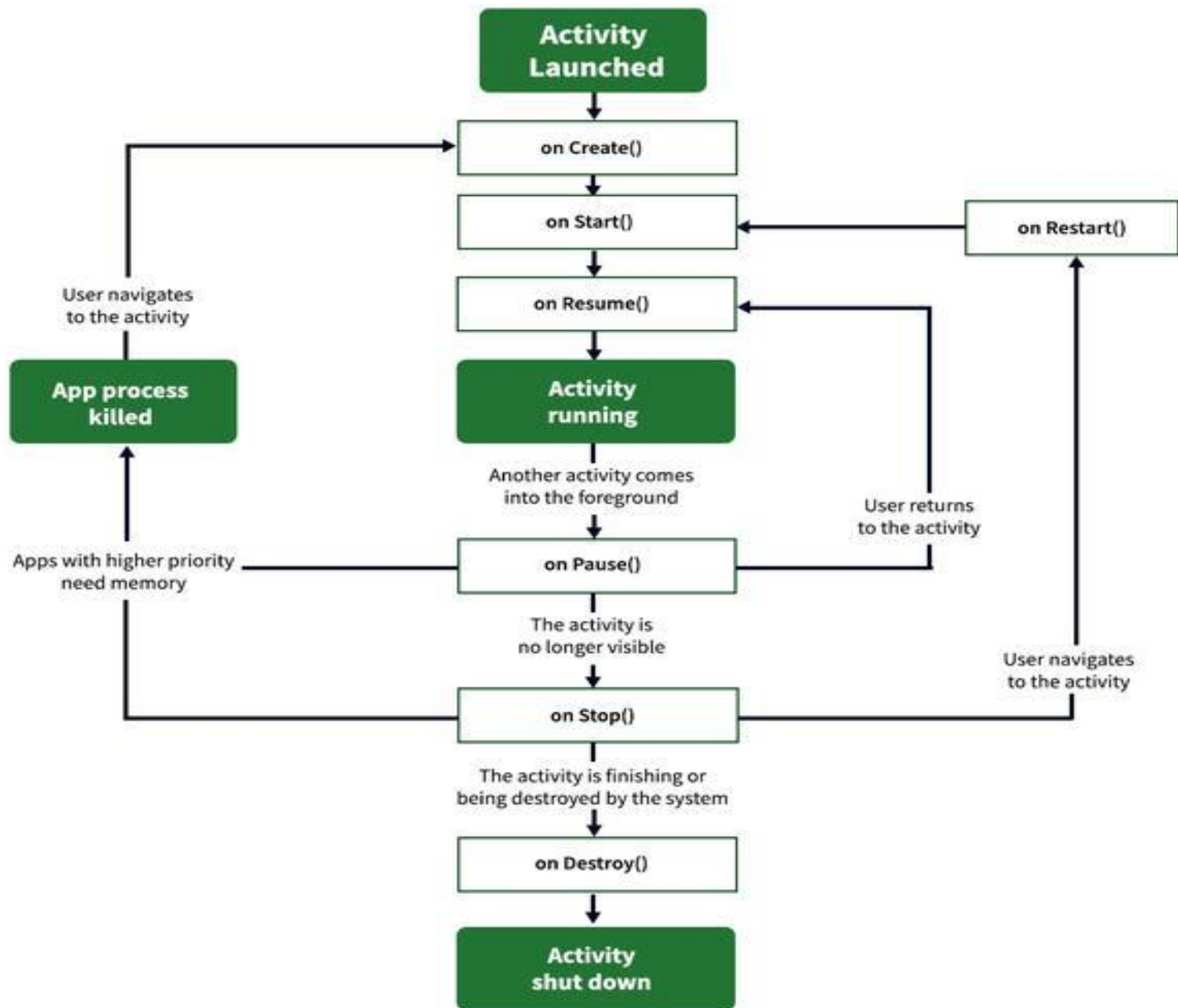


Activity and Fragments

Activity

- In Android, an **activity** is referred to as **one screen in an application**.
- It is very similar to a single window of any desktop application.
- An Android app consists of one or more screens or activities.
- Each activity goes through **various stages or a lifecycle** and is managed by activity stacks.
- So when a new activity starts, the previous one always remains below it.



Activity Lifecycle in Android

```
package com.example.myapplicationacactivitylifecycle;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        System.out.print("lifecycle onCreate invoked");
    }
    @Override
    protected void onStart() {
        super.onStart();
        Log.d("lifecycle", "onStart invoked");
        Toast.makeText(getApplicationContext(), "onStart Called",
        Toast.LENGTH_LONG).show();
    }
}
```

@Override

```
protected void onResume() {  
    super.onResume();  
    Log.d("lifecycle","onResume invoked");  
    Toast.makeText(getApplicationContext(), "onResume Called",  
        Toast.LENGTH_LONG).show();  
}
```

@Override

```
protected void onPause() {  
    super.onPause();  
    Log.d("lifecycle","onPause invoked");  
    Toast.makeText(getApplicationContext(), "onPause Called",  
        Toast.LENGTH_LONG).show();  
}
```

@Override

```
protected void onStop() {  
    super.onStop();  
    Log.d("lifecycle","onStop invoked");  
    Toast.makeText(getApplicationContext(), "onStop Called",  
        Toast.LENGTH_LONG).show();  
}
```

@Override

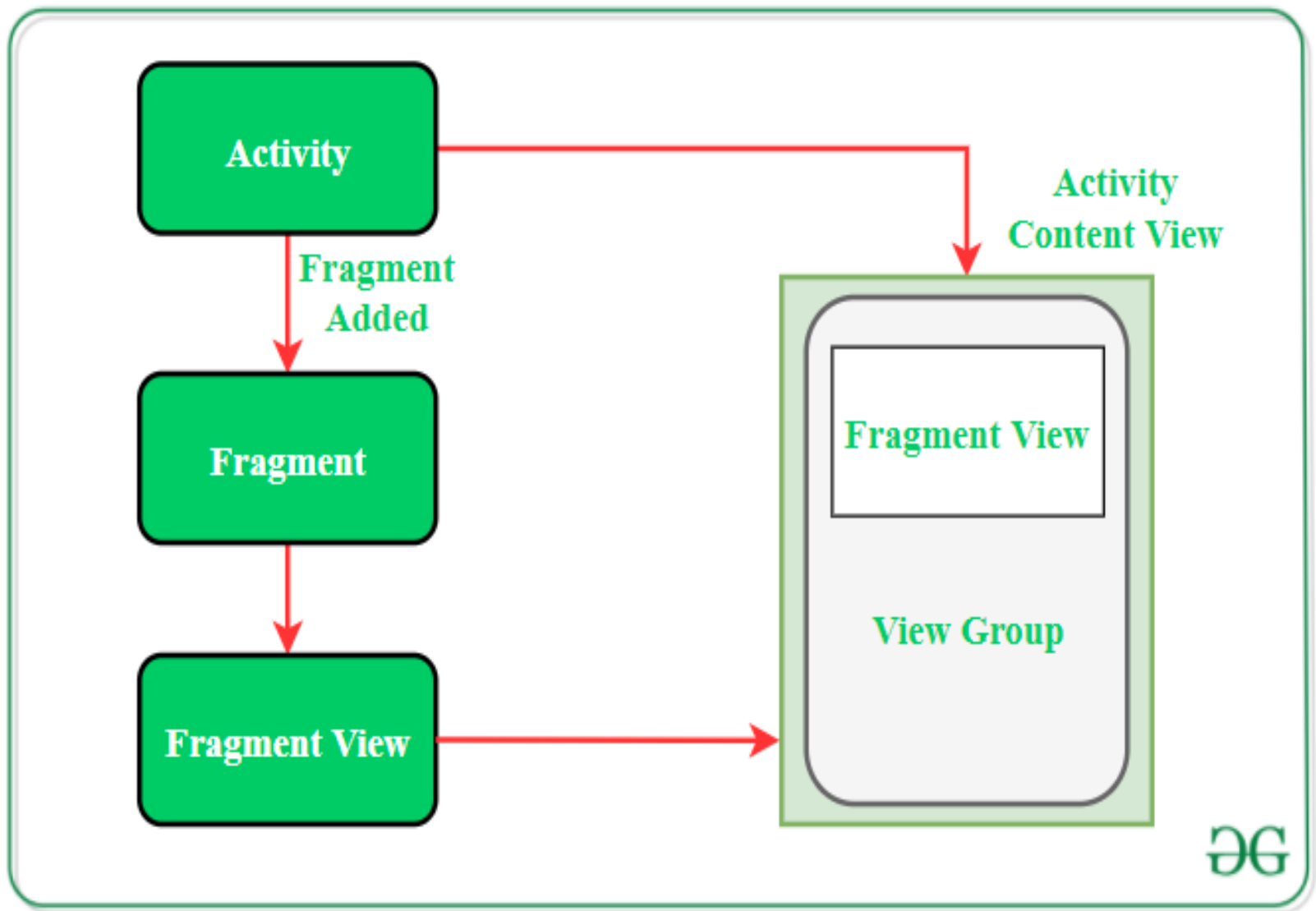
```
protected void onRestart() {  
    super.onRestart();  
    Log.d("lifecycle","onRestart invoked");  
    Toast.makeText(getApplicationContext(), "onRestart Called",  
        Toast.LENGTH_LONG).show();  
}
```

@Override

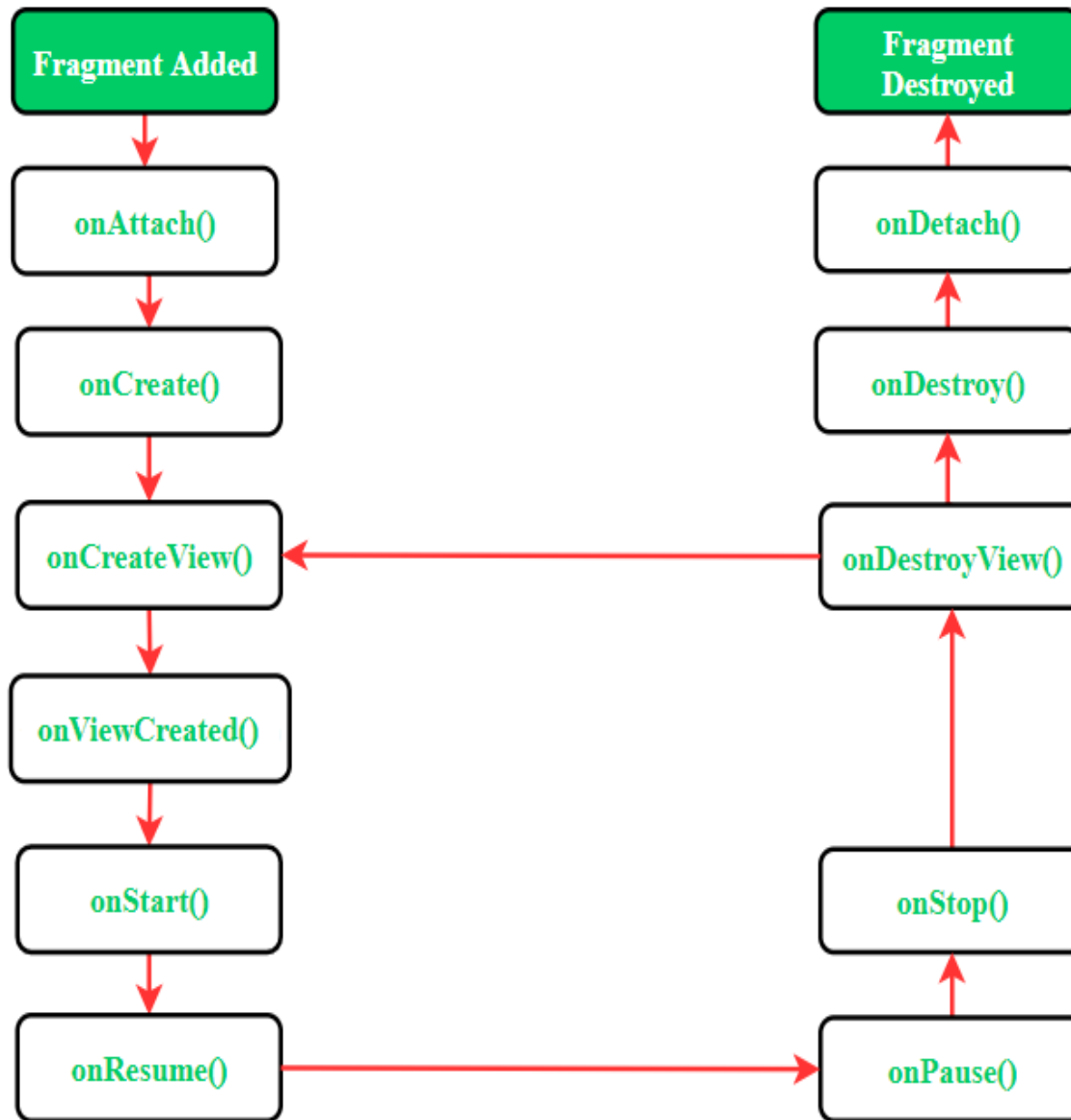
```
protected void onDestroy() {  
    super.onDestroy();  
  
    Log.d("lifecycle","onDestroy invoked");  
    Toast.makeText(getApplicationContext(), "onDestroy Called",  
        Toast.LENGTH_LONG).show();  
}  
}
```

Fragment

- In Android, the **fragment is the part of the Activity** that represents a **portion of the User Interface(UI)** on the screen.
- It is the modular section of the Android activity that is very helpful in creating UI designs that are flexible in nature and auto-adjustable based on the device screen size.
- The UI flexibility on all devices improves the user experience and adaptability of the application.
- An activity as its lifecycle is dependent on the lifecycle of the host activity.
- For example, if the host activity is paused, then all the methods and operations of the fragment related to that activity will stop functioning, the fragment is also termed a sub-activity.
- Fragments in Android can be added, removed, or replaced dynamically i.e., while the activity is running.



Fragment interaction with the activity



Fragment Life Cycle

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
<TextView
```

activity_main.xml

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Hello World!"/>
```

```
<fragment
```

```
    android:id="@+id/fragment1"
    android:name="com.example.myapplicationactivity.fragment1"
    android:layout_width="0px"
    android:layout_height="match_parent"
    android:layout_weight="1"
    android:background="#0000FF"
```

```
/>
```

```
</LinearLayout>
```

```
package com.example.myapplicationactivity;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

MainActivity.java

To add new fragment

File → New → Fragment → Fragment (with ViewModel) → give name as fragment1

fragment1_fragment.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".fragment1"
    android:background="#0000FF">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:text="Hello"/>

</FrameLayout>
```

